

Patrick Wagstrom

Data Engineering and Artificial Intelligence Leader

patrick@wagstrom.net, [Coventry, CT](#)

Strategic and hands-on executive leader in artificial intelligence, machine learning, data engineering, and software development, with experience across legal, finance, telecom, and video streaming. I drive cutting-edge AI/ML initiatives, build high-performing teams, modernize data pipelines, and deliver significant business value.

Professional Experience

Thomson Reuters — Hartford, CT (Remote)

Distinguished Engineer, January 2025 - Present

- Led engineering for development of Westlaw Advantage - our legal Deep Research solution and most successful product launch ever.
- Led overall work on initial CoCounsel Legal MCP server and our integration with Anthropic and Claude Legal as their first external plugin.
- Led engineering for Westlaw Brief Builder - our agentic solution that tackles some of the hardest legal challenges - drafting litigation briefs.
- Rationalized complex authorization, authentication, and integration with legacy systems and updating for an autonomous agentic driven world.
- Bring together engineers and scientists weekly to lead culture around agentic development with Agentic Coffeehouse series.
- Work directly with scientists and frontier AI labs on design, implementation, and evaluation of future model and agentic solutions.

Grainger — Chicago, IL and Hartford, CT (Hybrid)

Senior Director Applied ML, July 2023 - August 2024

- Led team of 43 FTE ML scientists, software engineers, and product managers on product search and discovery, LLM agents, computer vision, ML in mobile applications, data pipelines, and governance.
- Developed an LLM-based customer service agent using search and unstructured data with OpenAI on Azure to provide human-in-the-middle assistance to Grainger employees.
- Enhanced the search capabilities of Grainger.com, the 11th largest ecommerce site in the US, through introduction of vector and hybrid search.
- Led team that built out innovative synthetic data generation product identification using 3D CAD models, Stable Diffusion, and Unity.
- Designed the overall data architecture and MLOps integration - an S3 data lake with data products, Kafka event streaming, Databricks, Snowflake, and Atlan for data governance.
- Defined the ML Scientist job family across the company, integrated it into overall talent strategy, and partnered with legal to implement generative AI governance.

Brightcove — Boston, MA and Hartford, CT (Hybrid)

Chief Data Officer, May 2021 - December 2022

- Responsible for all analytics, data, and machine learning at Brightcove - including engineering, governance, policy, and vendor relations.
- Owned end-to-end analytics collection and processing systems handling trillions of rows of data and billions of monthly video views on both AWS and GCP.
- Managed an organization of 34 FTEs and 12 contractors, overseeing data platforms, analytics and insights, data and model governance, and customer-facing data products.
- Executed the \$12.3 million acquisition of Wicket Labs, expanding Brightcove's viewer analytics capabilities and growing the product customer base by over 50x in under a year.

- Negotiated and owned a multi-million dollar partnership with Google Cloud Platform - securing a 7-figure discount.
- Maintained and expanded multi-cloud data infrastructure - Trino, Glue, Lambda, SageMaker, and Redshift on AWS; BigQuery, Composer, Dataflow, and Pub/Sub on GCP.

Verizon — Basking Ridge, NJ and Hartford, CT (Hybrid)

Director of Emerging Technology, September 2019 - April 2021

- Built and managed a team of 9 FTEs and 51 vendor contractors with a \$15 million annual budget, collaborating closely with CIO, CTO, and CDAO to influence corporate strategies.
- Worked alongside legal and policy experts to create several Verizon policies, including facial recognition, energy efficiency, model risk management, and data privacy.
- Developed cell site energy efficiency machine learning models, resulting in \$5 million annual savings and the creation of a digital twin for future simulation.
- Led development of a model connecting online brand discussions to customer profiles, supporting customer service and patent-pending bot detection.
- Designed a new data architecture to migrate from ad-hoc on-prem Teradata and Hadoop solutions to a managed streaming data architecture with Google BigQuery.

Capital One — New York, NY and McLean, VA

Senior Director of Data Science, January 2019 - September 2019

Director of Data Science, November 2016 - January 2019

- Architect and overall lead of the Capital One Card Machine Learning Platform - scalable, personalized, real-time reinforcement learning models on Kubernetes, gRPC, and AWS in a heavily regulated environment.
- Led a team that deployed and managed multi-modal customer intelligence models working on both structured and unstructured text and voice data, resulting in \$27 million annual savings.
- Hired, managed, and developed a distributed team of 18 data scientists and data analysts.
- Championed reproducible machine learning and MLOps across the organization, ensuring best practices in data management, model training and refit, model serving, and risk management.

IBM Research — Littleton, MA and Yorktown Heights, NY

Research Staff Member/Technical Lead, January 2014 - November 2016

Research Staff Member, August 2009 - January 2014

- Led a globally distributed team that built the IBM Watson Conversation service - creating rich conversational interfaces with natural language processing.
- Global team lead for Watson Developer Cloud Tooling - applications to create, train, and maintain cognitive and ML solutions including Watson Engagement Advisor and Natural Language Classifier.
- Technical lead for Chef Watson and the IBM Food Truck - one of the first consumer facing generative AI solutions - garnering more than 1 billion media impressions in 2014.
- Researched, designed, and built tools to customize AI/ML models at scale for hundreds of thousands of users.
- Team lead for the "Millennial Enterprise" chapter of IBM's 2013 Global Technology Outlook, presented to IBM and customer C-Level executives.

Education

Ph.D. in Engineering and Public Policy and Computation, Organizations, and Society, May 2009

Carnegie Mellon University, Pittsburgh, PA

Thesis: "Vertical Interaction in Open Software Engineering Communities". Advisors: Dr. James Herbsleb and Dr. Kathleen Carley.

MS in Computation, Organizations, and Society, May 2007

Carnegie Mellon University, Pittsburgh, PA

MS in Computer Science, August 2003

Illinois Institute of Technology, Chicago, IL

BS in Computer Science / BS in Computer Engineering / BS in Electrical Engineering, May 2002

Illinois Institute of Technology, Chicago, IL